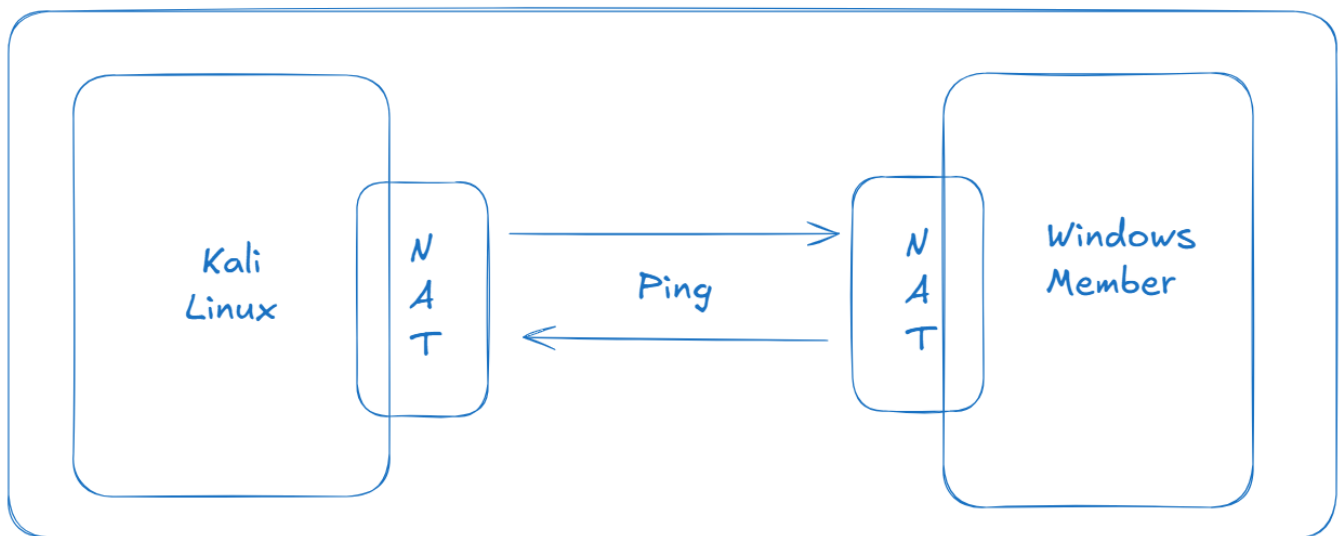


- Students to analyse the given malware analysis link and answer the questions
 - Lab setup:
 - 1st VM: Kali linux
 - Tools needed:
 - strings
 - sha256sum
 - tcpdump
 - python3 -m http.server
 - 2nd VM: Windows Server 2022 (member)
 - Tools needed:
 - Windows Defender
 - PowerShell
 - Event Viewer

Labsetup:

VMWare Workstation



On kali linux:

Create a Test "Malware" File (EICAR - European Institute for Computer Anti-virus Research (EICAR))

- `mkdir ~/malware-demo`
- `cd ~/malware-demo`
- `echo 'X5O!P%@AP[4\PZX54(P^)7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H*' > eicar.txt`
 - This is 68-bytes string of ASCII text.
 - This is a fixed string.
 - EICAR is used for testing anti-virus software functionality without using real Malware.

Verify hash:

- `sha256sum eicar.txt`

Start a Local Web Server (Delivery Simulation)

- `python3 -m http.server 8080`

```
(root@kali)-[~]
# mkdir ~/malware-demo
cd ~/malware-demo

echo 'X5O!P%@AP[4\PZX54(P^)7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H*' > eicar.txt

(root@kali)-[~/malware-demo]
# ls
eicar.txt

(root@kali)-[~/malware-demo]
# sha256sum eicar.txt
131f95c51cc819465fa1797f6ccacf9d494aaaff46fa3eac73ae63ffbfd8267 eicar.txt

(root@kali)-[~/malware-demo]
# python3 -m http.server 8080

Serving HTTP on 0.0.0.0 port 8080 (http://0.0.0.0:8080/) ...
192.168.113.141 - - [31/Jan/2026 13:14:43] "GET /eicar.txt HTTP/1.1" 200 -
192.168.113.141 - - [31/Jan/2026 13:16:51] "GET /eicar.txt HTTP/1.1" 200 -
```

On windows:

Run below commands on PowerShell as Administrator.

- Set-MpPreference -DisableRealtimeMonitoring \$false
- netsh advfirewall set allprofiles state on

Verify Network Connectivity

- ping <KALI_IP>

Download the File (Simulated Infection Vector)

- Invoke-WebRequest http://<KALI_IP>:8080/eicar.txt -OutFile C:\Users\Administrator\Desktop\ecar.txt

Event Logs

Now go to Event Viewer → Application and Service logs → Microsoft → Windows → Windows Defender

Level	Date and Time	Source	Event ID	Task Category
Information	1/31/2026 10:17:02 AM	Windows Defender	1117	None
Warning	1/31/2026 10:16:51 AM	Windows Defender	1116	None
Information	1/31/2026 10:16:32 AM	Windows Defender	1117	None
Warning	1/31/2026 10:16:20 AM	Windows Defender	1116	None
Information	1/31/2026 10:16:15 AM	Windows Defender	5000	None
Information	1/31/2026 1:13:44 AM	Windows Defender	1117	None
Information	1/31/2026 1:13:44 AM	Windows Defender	1117	None
Information	1/31/2026 1:13:44 AM	Windows Defender	1117	None
Information	1/31/2026 1:13:34 AM	Windows Defender	5007	None
Information	1/31/2026 1:13:34 AM	Windows Defender	1001	None
Warning	1/31/2026 1:13:34 AM	Windows Defender	1116	None
Warning	1/31/2026 1:13:34 AM	Windows Defender	1116	None
Warning	1/31/2026 1:13:34 AM	Windows Defender	1116	None
Information	1/31/2026 1:13:34 AM	Windows Defender	2010	None
Information	1/31/2026 1:13:34 AM	Windows Defender	2010	None
Information	1/31/2026 1:12:48 AM	Windows Defender	5007	None
Information	1/31/2026 1:12:45 AM	Windows Defender	1013	None
Information	1/31/2026 1:12:44 AM	Windows Defender	1000	None
Error	1/31/2026 1:06:33 AM	Windows Defender	1119	None
Warning	1/31/2026 1:06:27 AM	Windows Defender	1116	None
Error	1/31/2026 1:06:27 AM	Windows Defender	1119	None
Information	1/31/2026 1:06:27 AM	Windows Defender	5001	None

Event 1117, Windows Defender

General Details

Windows Defender has taken action to protect this machine from malware or other potentially unwanted software. For more information please see the following:
https://go.microsoft.com/fwlink/?linkid=37020&name=Virus:DOS/EICAR_Test_File&threatid=2147519003&enterprise=0

Log Name: Microsoft-Windows-Windows Defender/Operational
Source: Windows Defender Logged: 1/31/2026 10:17:02 AM
Event ID: 1117 Task Category: None
Level: Information Keywords:
User: SYSTEM Computer: WIN-SAFJH31R8E8
OpCode: Info
More Information: [Event Log Online Help](#)

Look for:

- Event ID 1116 / 1117

Capture Traffic:

Run as root on kali:

- `tcpdump -i eth0 port 8080 -n`

Now, go to window server and rerun the same command:

- Invoke-WebRequest `http://<KALI_IP>:8080/eicar.txt -OutFile C:\Users\Public\ecar.txt`

Go to kali linux again, rerun the tcpdump command:

- `tcpdump -i eth0 port 8080 -n`

```
(root@kali)~#  
# tcpdump -i eth0 port 8080 -n  
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode  
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes  
13:33:35.829366 IP 192.168.113.141.49708 > 192.168.113.136.8080: Flags [SEW], seq 130062625, win 8192, options  
0  
13:33:35.829409 IP 192.168.113.136.8080 > 192.168.113.141.49708: Flags [S.], seq 706269845, ack 130062626, win  
cale 9], length 0  
13:33:35.829733 IP 192.168.113.141.49708 > 192.168.113.136.8080: Flags [P.], seq 1:174, ack 1, win 2053, length 0  
13:33:35.829934 IP 192.168.113.141.49708 > 192.168.113.136.8080: Flags [P.], seq 1:174, ack 1, win 2053, length 0  
13:33:35.829942 IP 192.168.113.136.8080 > 192.168.113.141.49708: Flags [P.], seq 1:188, ack 174, win 126, length 0  
13:33:35.830781 IP 192.168.113.136.8080 > 192.168.113.141.49708: Flags [P.], seq 1:188, ack 174, win 126, length 0  
13:33:35.830921 IP 192.168.113.136.8080 > 192.168.113.141.49708: Flags [FP.], seq 188:257, ack 174, win 126, length 0  
13:33:35.831122 IP 192.168.113.141.49708 > 192.168.113.136.8080: Flags [P.], seq 1:188, ack 174, win 126, length 0  
13:33:35.858011 IP 192.168.113.141.49708 > 192.168.113.136.8080: Flags [F.], seq 174, ack 258, win 2052, length 0  
13:33:35.858032 IP 192.168.113.136.8080 > 192.168.113.141.49708: Flags [P.], seq 174, ack 258, win 2052, length 0
```

Static Analysis demo (on kali)

```
(root@kali)~#  
# ls  
ecar.txt  
  
(root@kali)~#  
# strings eicar.txt  
X50!P%AP[4\PZX54(P^)7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H*  
  
(root@kali)~#  
# cat eicar.txt  
X50!P%AP[4\PZX54(P^)7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H*  
  
(root@kali)~#  
# file eicar.txt  
ecar.txt: EICAR virus test files  
  
(root@kali)~#
```

- Students will learn to create phishing links → done already.